Financial Mathematics 32000

Lecture 2

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UNIT 1: Trees

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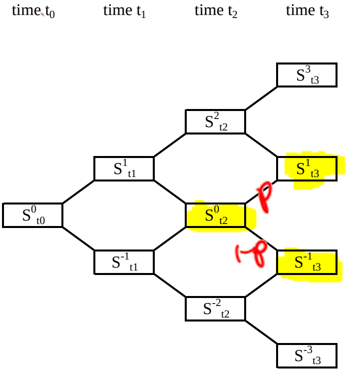
Pricing options; Europeans, barriers, Americans

Approximating diffusions: GBM and Local Volatility

Appendix: Lookback option

Binomial tree

Consider an N-period model (example: $N = 3$) with a bank account $B_t = e^{rt}$ and a stock S with dynamics

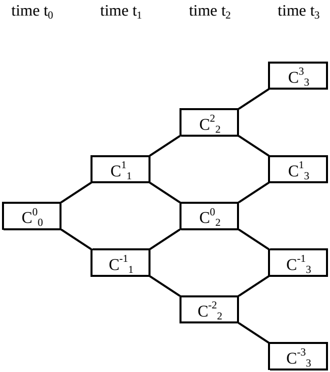


Superscripts are not powers. Subscripts $t_n = n\Delta t$ for $n = 0, \ldots, N$.

Binomial tree

Want to find time-0 price C_0^0 of some option.

To simplify notation, we will write C_n^j instead of $C_{t_n}^j$.



Consider three contract types: European, barrier, and American.

Pricing a European

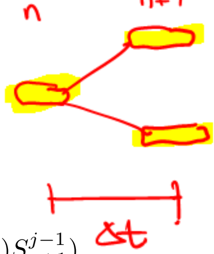
If the expiry date is at time $T = t_N$ and the payoff function is $f(S)$, then for all j, the option price is

$f(S) = (S-K)^+$

if call

$$C_N^j = f(S_{t_N}^j).$$

Now induct backwards. If C_{n+1}^j has been computed for all j, then



$$C_n^j = e^{-r\Delta t}[p_n^j C_{n+1}^{j+1} + (1 - p_n^j) C_{n+1}^{j-1}],$$

where

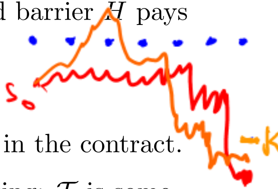
$$p_n^j := \frac{S_{t_n}^j e^{r\Delta t} - S_{t_{n+1}}^{j-1}}{S_{t_{n+1}}^{j+1} - S_{t_{n+1}}^{j-1}},$$

either by replication, or by $S_n^j = e^{-r\Delta t}(p_n^j S_{n+1}^{j+1} + (1 - p_n^j) S_{n+1}^{j-1})$.

Pricing an up-and-out barrier option

An *up-and-out put* on S with strike K, expiry T, and barrier H pays

$$(K - S_T)^+ \mathbf{1}_{\max_{t\in\mathcal{T}} S_t < H}.$$



where $\mathcal{T} \subseteq [0, T]$ are the observation times, specified in the contract. Continuous monitoring: $\mathcal{T} = [0, T]$. Discrete monitoring: $\mathcal{T}$ is some finite set. The option price is a function of *three* state variables:

$$C(S^j_{t_n},\ t_n,\ \mathbf{1}_{\text{knockout prior to time } t_n}).$$

where "knockout prior to time t_n" is the event that there exists $t < t_n$ with $t \in \mathcal{T}$ and $S_t \geq H$. Computationally: $C(S, t, 1)$ does not need to be tracked; it's always zero. So what we track in the tree is $C^j_n := C(S^j_{t_n}, t_n, 0)$, the option price *in the case of no prior knockout.*

Pricing an up-and-out barrier option

Let $t_N = T$. At terminal nodes, $C_N^j = (K - S_{t_N}^j)^+ \mathbf{1}_{S_{t_N}^j < H \text{ or } t_N \notin \mathcal{T}}$.

Inducting backwards, if C_{n+1}^j has been computed for all j, then at node (j, n) we have two cases:

$$t_n \in \mathcal{T} \text{ and } S_{t_n}^j \geq H \Rightarrow \quad C_n^j = 0 \tag{1}$$

$$t_n \notin \mathcal{T} \text{ or } S_{t_n}^j < H \Rightarrow \quad C_n^j = e^{-r\Delta t}[p_n^j C_{n+1}^{j+1} + (1 - p_n^j) C_{n+1}^{j-1}] \tag{2}$$

So everywhere at or beyond the barrier at monitoring times,

set the price to 0; elsewhere, take the usual discounted average of the next-time-step values.

(And in building a tree, you try to have the dates important in the contract – such as expiry and barrier monitoring dates – be among the dates represented in the tree.)

Pricing an American put

Notation: $x \wedge y := \min(x, y)$.

On a time interval $[t_1, t_2 \wedge \tau]$, consider a portfolio with price process V, that includes a static position in an American option, to be exercised at time $\tau \geq t_1$. We say it is an *arbitrage* if $V_{t_1} = 0$ and

$$\mathsf{P}(V_{t_2 \wedge \tau} < 0) = 0 \text{ and } \mathsf{P}(V_{t_2 \wedge \tau} > 0) > 0$$

for *some* stopping time $\tau \geq t_1$, if the portfolio is long the option;
for *all* stopping times $\tau \geq t_1$, if the portfolio is short the option.

Pricing an American

At expiry $T = t_N$, we have $C_N^j = (K - S_{t_N}^j)^+$. Inducting backwards, if C_{n+1}^j is the no-arb price that has been computed for each j then

$$C_n^j = \max((K - S_{t_n}^j)^+,\ e^{-r\Delta t}[p_n^j C_{n+1}^{j+1} + (1 - p_n^j) C_{n+1}^{j-1}])$$

This is because the holder at node (j, n) can either exercise and receive $(K - S_{t_n}^j)^+$, or hold on to the option which tomorrow will have no-arbitrage price of C_{n+1}^{j+1} (if up) or C_{n+1}^{j-1} (if down), which implies

- If $C_n^j < \max$, then there is arbitrage: go long the option.
- If $C_n^j > \max$, then there is arbitrage: short the option; if holder exercises, you close out with a profit; if not, then use the funds to buy a portfolio that superreplicates the time-t_{n+1} option value.

The time-0 option price is C_0^0.

Pricing an American: another formulation

Math fact: For any adapted process Z_{t_n}, define $Y_{t_N} = Z_{t_N}$ and

$$Y_{t_n} = \max(Z_{t_n},\ \mathbb{E}_{t_n} Y_{t_{n+1}}), \qquad n = N-1, N-2, \ldots, 0.$$

(E.g.: $Z_t = e^{-rt}(K - S_t)^+$ and $Y_t = e^{-rt} \times$ time-t value of American).

Then the **optimality principle of dynamic programming** says that

$$Y_0 = \max\{\mathbb{E} Z_\tau :\ \tau \text{ is a stopping time, } 0 \le \tau \le T\}$$

and $Y_0 = \mathbb{E} Z_{\tau^*}$ where $\tau^* := \min\{t_n :\ Y_{t_n} = Z_{t_n}\}$.

This leads to another formulation of the American option price:

$$C_0 = \max\{\mathbb{E} e^{-r\tau}(K - S_\tau)^+ :\ \tau \text{ is a stopping time, } 0 \le \tau \le T\}$$

(which is also valid in continuous time for continuous processes S).

Another application of dynamic programming

Interview (?!) question:

Take a 52 card deck with 26 red and 26 black cards, in random order.

Reveal cards sequentially without replacement.

With each black card you get $+1$ dollar.

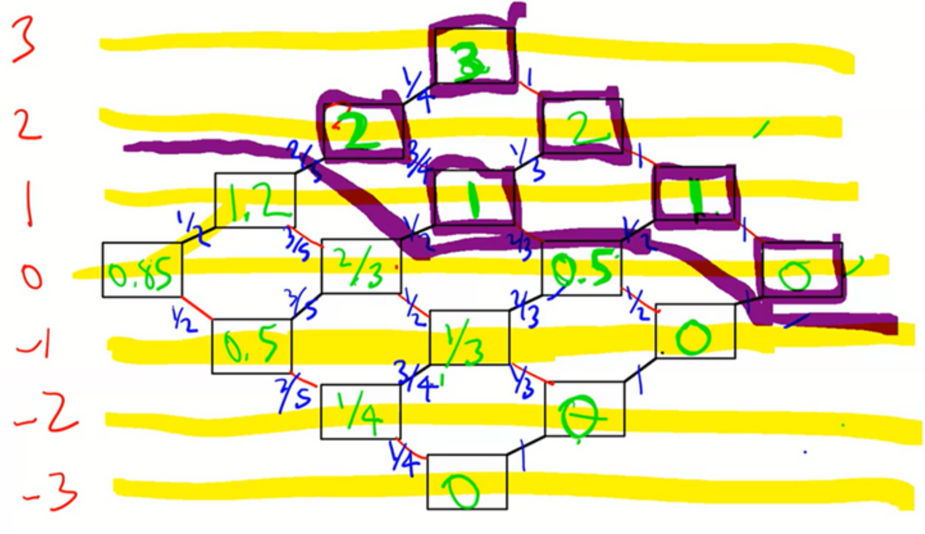
With each red card you get -1 dollar.

You can stop playing the game at any time.

What's the optimal stopping time, and what's your expected profit from the optimal strategy?

(Easier question: 6 cards – 3 red, 3 black)

Another application of dynamic programming



UNIT 1: Trees

Pricing options; Europeans, barriers, Americans

Approximating diffusions: GBM and Local Volatility

Appendix: Lookback option

Trees as approximations of diffusions

- Approximate the risk-neutral dynamics of a diffusion process (such as GBM) using a discrete model with a finite number of branches from each state.
- The expectations (hence option prices) that we find in the tree approximate the expectations (hence option prices) under the diffusion process.
- For example, let us use a tree to approximate the diffusion

$$\mathrm{d}S_t = R_{grow} S_t \mathrm{d}t + \sigma S_t \mathrm{d}W_t$$

Use binomial tree to approximate diffusion dynamics

- Diffusion is

$$dS_t = R_{grow} S_t dt + \sigma S_t dW_t$$

where W is $\mathbb{P}$-BM. So $X := \log S$ has dynamics

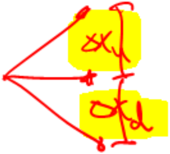
$$dX_t = \nu dt + \sigma dW_t$$

where $\nu := R_{grow} - \sigma^2/2$.

- Approximate dynamics of $X := \log S$ using a binomial tree.

$$\mathbb{P}(X_{t+\Delta t} = X_t + \Delta x_u) = p$$

$$\mathbb{P}(X_{t+\Delta t} = X_t - \Delta x_d) = 1 - p$$



where Δx_u and Δx_d and p will be nonrandom parameters.

Match the means and variances

- Let $\Delta t := T/N$ where N is number of time steps.
- Now choose $p, \Delta x_u, \Delta x_d$ such that diffusion and tree agree on mean and variance of the increments. In the diffusion,

$$X_{t+\Delta t} - X_t = \int_t^{t+\Delta t} \nu \mathrm{d}s + \int_t^{t+\Delta t} \sigma \mathrm{d}W_s = \nu \Delta t + \sigma \Delta W.$$

So

$$\nu \Delta t = \mathbb{E}_t(X_{t+\Delta t} - X_t) = p\Delta x_u + (1-p)(-\Delta x_d)$$

$$\sigma^2 \Delta t = \mathrm{Var}_t(X_{t+\Delta t} - X_t) = p(\Delta x_u)^2 + (1-p)(-\Delta x_d)^2 - (\nu \Delta t)^2$$

Two equations and three unknowns: p, Δx_u, Δx_d. Can impose another condition (such as $\Delta x_u = \Delta x_d$ or $p = 1/2$) and solve.

Then price the option in a tree with these parameters.

Not flexible enough

- However, often we want to specify Δx_u and Δx_d in advance.

  For example, we may want $\Delta x_u = \Delta x_d = \Delta x$, so that the *same* price levels are represented at every time point in the tree.

  Maybe, furthermore, we want to specify the size of Δx to optimize convergence or to choose *which* price levels are represented.

- This leaves only one free parameter p.

  Not enough to match both the mean and the variance.

Solution: trinomial trees.

Use trinomial tree to approximate diffusion dynamics

- Diffusion is

$$\mathrm{d}S_t = R_{grow} S_t \mathrm{d}t + \sigma S_t \mathrm{d}W_t$$

where W is $\mathbb{P}$-BM. So $X := \log S$ has dynamics

$$\mathrm{d}X_t = \nu \mathrm{d}t + \sigma \mathrm{d}W_t$$

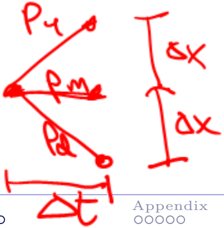
where $\nu := R_{grow} - \sigma^2/2$.

- Approximate dynamics of $X := \log S$ using a trinomial tree.

$$\mathbb{P}(X_{t+\Delta t} = X_t + \Delta x) = p_u$$

$$\mathbb{P}(X_{t+\Delta t} = X_t) = p_m$$

$$\mathbb{P}(X_{t+\Delta t} = X_t - \Delta x) = p_d$$



Match the means and variances

- Let $\Delta t := T/N$ where N is number of time steps.

  Choose $\Delta x \approx \sigma\sqrt{3\Delta t}$ for accuracy reasons.

  Can modify these suggestions to make sure certain price levels or times are represented in the tree.

- Now choose p_u, p_m, p_d such that diffusion and tree agree on mean and variance of the increments. In the diffusion,

$$X_{t+\Delta t} - X_t = \int_t^{t+\Delta t} \nu \mathrm{d}s + \int_t^{t+\Delta t} \sigma \mathrm{d}W_s = \nu\Delta t + \sigma\Delta W.$$

  So

$$\nu\Delta t = \mathbb{E}_t(X_{t+\Delta t} - X_t) = p_u\Delta x + p_m 0 + p_d(-\Delta x)$$

$$\sigma^2\Delta t = \mathrm{Var}_t(X_{t+\Delta t} - X_t) = p_u(\Delta x)^2 + p_m(0)^2 + p_d(-\Delta x)^2 - (\nu\Delta t)^2$$

$$p_u + p_m + p_d = 1$$

Solve for probabilities

Solve system of three equations in three unknowns.

$$
\begin{aligned}
p_u &= \frac{1}{2}\left[\frac{\sigma^2\Delta t + \nu^2(\Delta t)^2}{(\Delta x)^2} + \frac{\nu\Delta t}{\Delta x}\right] \\
p_m &= 1 - \frac{\sigma^2\Delta t + \nu^2(\Delta t)^2}{(\Delta x)^2} \\
p_d &= \frac{1}{2}\left[\frac{\sigma^2\Delta t + \nu^2(\Delta t)^2}{(\Delta x)^2} - \frac{\nu\Delta t}{\Delta x}\right]
\end{aligned}
$$

Intuition: For fixed Δt and Δx,

- The bigger the σ, the more probability mass in the wings, the less in the middle.
- The bigger the ν, the more mass in the up-branch, the less in the down-branch.

Option pricing

To find option prices, induct backwards in tree with time points

$$t_n = n\Delta t \qquad n = 0, \ldots, N,$$

and log-stock-price points

$$x_j = \log S_0 + j\Delta x \qquad j = -N, \ldots, N.$$

Option price in tree for node at time t_n and log-price x_j is

$$C_n^j = e^{-r\Delta t}[p_u C_{n+1}^{j+1} + p_m C_{n+1}^j + p_d C_{n+1}^{j-1}].$$

With large enough N, the option price in the tree $\approx$ the option price for diffusion, because in the $N \to \infty$ (or equivalently $\Delta t \to 0$) limit, the tree's option price $\to$ option price under diffusion dynamics.

For Europeans, the proof is by a form of the CLT.

Can't replicate general payoffs if 3 states 2 assets

- Recall that in the 3-state model, general options cannot be replicated using stock and bank acct. There is not a unique arbitrage-free price for the option. So how can we talk about finding "the" price of the option?
- Answer: The trinomial tree is not the model of the market.
  The model is GBM in continuous time. In that model, replication using $\{B, S\}$ is possible. So unique price exists.
  The tree is a *computational device* that we use in order to approximate the risk-neutral expectations (hence the prices) of the continuous-time model.

Option Pricing: Knock-outs and Americans

Pricing of knock-outs

$$C_n^j = e^{-r\Delta t}[p_u C_{n+1}^{j+1} + p_m C_{n+1}^{j} + p_d C_{n+1}^{j-1}]\mathbf{1}_{\text{no knock-out at time } t_n}$$

Pricing of American put

$$C_n^j = \max((K - S_{t_n}^j)^+,\ e^{-r\Delta t}[p_u C_{n+1}^{j+1} + p_m C_{n+1}^{j} + p_d C_{n+1}^{j-1}])$$

Formulas are similar to binomial tree.

Local volatility models

We want a model that is consistent with the observed non-constant implied vol skew. One approach: *local volatility* models specify the instantaneous volatility σ to be a function of (S_t, t).

- In the continuous-time setting,

$$\mathrm{d}S_t = rS_t\mathrm{d}t + \sigma(S_t, t)S_t\mathrm{d}W_t$$

  where W is $\mathbb{P}$-BM.

- In the tree setting, we let σ depend on the node (S_t, t).

Note that the local volatility σ is not the same thing as σ_{imp}.

Analytic computation of option prices is difficult in the diffusion setting, even for European calls/puts (exception: $\sigma(S_t, t) = \sigma(t)$). But by approximating the diffusion in a tree, the computations are easy.

Option pricing in the tree setting

Suppose we are given the local volatility function σ.

- Let $\Delta t = T/N$ (unless this fails to place important dates in tree).
- To choose Δx, let σ_{avg} be some "representative" or "average" σ in the tree, and let σ_{max} be an upper bound on the σ in the tree. Two guidelines: to make local discretization error small, we want

$$\Delta x \approx \sigma_{avg}\sqrt{3\Delta t}$$

but for stability reasons, we want

$$\Delta x \geq \sigma_{max}\sqrt{\Delta t}$$

So we can let $\Delta x = \max(\sigma_{avg}\sqrt{3\Delta t}, \sigma_{max}\sqrt{\Delta t})$

Option pricing in the tree setting

- Then at each node, use the $\sigma(S_t, t)$ prevailing at that particular node to generate the ν and the probabilities for the branches out of that node. Same formulas as L1:

$$p_{u,d} = \frac{1}{2}\left[\frac{\sigma^2\Delta t + \nu^2(\Delta t)^2}{(\Delta x)^2} \pm \frac{\nu\Delta t}{\Delta x}\right], \quad p_m = 1 - \frac{\sigma^2\Delta t + \nu^2(\Delta t)^2}{(\Delta x)^2}$$

- Price options – including path-dependent and American-style options – as we did for GBM.

  The only change is that σ (hence ν, p_u, p_m, p_d) vary across nodes.

What if σ is not given

Then *calibrate* it to the prices of listed options.

The general idea of calibration:

- Observe prices of liquidly traded assets (e.g. listed options)
- Choose the model's parameters (e.g. the σ function) in such a way that the model generates theoretical prices that match closely the observed prices.

Then one can apply that model, with the calibrated parameters, to

- Compute hedges and risk sensitivities
- Price and hedge illiquid options
- Price and hedge complex contracts (e.g. structured products and exotic options)

Calibration of local volatility $\sigma(t)$

If σ is a non-random function of t and

$$\mathrm{d}S_t = rS_t\mathrm{d}t + \sigma(t)S_t\mathrm{d}W_t$$

then $\mathrm{d}\log S_t = (r - \frac{1}{2}\sigma^2(t))\mathrm{d}t + \sigma(t)\mathrm{d}W_t$ so

$$\log S_T = \log S_0 + \left(r - \frac{\bar{\sigma}_T^2}{2}\right)T + \int_o^T \sigma(t)\mathrm{d}W_t$$

$$\sim \text{Normal}\left(\log S_0 + (r - \frac{\bar{\sigma}_T^2}{2})T,\ \bar{\sigma}_T^2 T\right) \text{ where } \boxed{\bar{\sigma}_T := \sqrt{\frac{1}{T}\int_0^T \sigma^2(t)\mathrm{d}t}}$$

(intuition: a nonrandomly weighted sum of indep normals is normal)

So time-0 call prices $C(K,T) = C^{BS}(\bar{\sigma}_T)$, thus $\sigma_{imp}(K,T) = \bar{\sigma}_T$.

Calibration: If given $C(K,T)$ at various expiries T, then obtain σ_{imp} and use boxed equation to (not uniquely) find $\sigma(t)$.

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Calibration of local volatility $\sigma(S,t)$ in trinom tree

It's harder when σ depends on (S,t).

Here's the rough idea. Assume $r=0$ and a tree with S-levels s_j with equal spacing $\Delta S = s_{j+1} - s_j$. Let $\Delta K := \Delta S$.

Given: $C_0(K,\tau)$, the time-0 price of a strike-K expiry-τ call, for all (K,τ). Find at each node (s_j, t_n): Local volatility σ which generates risk-neutral probabilities consistent with call prices (and stock prices).

- The probability of reaching node (s_j, t_n) equals $(\Delta K)\mathrm{Fly}_0(s_j, t_n)$ where $\mathrm{Fly}_0(K,\tau)$ is the time-0 price of a *butterfly*:

$$\mathrm{Fly}_0(K,\tau) := \frac{C_0(K-\Delta K,\tau) - 2C_0(K,\tau) + C_0(K+\Delta K,\tau)}{(\Delta K)^2}$$

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Calibration of local volatility in a trinomial tree

- The probability of reaching node (s_j, t_n) *and then going up* equals $(\Delta t/\Delta K)\mathrm{Cal}_0(s_j, t_n)$, where $\mathrm{Cal}_0(K, \tau)$ is the time-0 price of a *calendar spread*:

$$\mathrm{Cal}_0(K,\tau) := \frac{C_0(K,\tau+\Delta t) - C_0(K,\tau)}{\Delta t}$$

by a replication argument.

Hence the conditional up-probability from node (s_j, t_n) is the ratio

$$p_u = \frac{\mathbb{P}(\text{reach } (s_j,t_n) \text{ then up})}{\mathbb{P}(\text{reach } (s_j,t_n))} = \frac{(\Delta t/\Delta K)\mathrm{Cal}_0}{(\Delta K)\mathrm{Fly}_0}$$

Calibration of local volatility in a trinomial tree

On the other hand, the up-probability is also obtained by solving

$$\begin{aligned} p_u(\Delta K) + p_m(0) + p_d(-\Delta K) &= 0 \\ p_u(\Delta K)^2 + p_m(0) + p_d(-\Delta K)^2 &= \sigma^2 s_j^2(\Delta t) \end{aligned} \quad \Rightarrow p_u = \frac{\sigma^2 s_j^2 \Delta t}{2(\Delta K)^2}$$

where σ, p_u, p_m, p_d all depend on (s_j, t_n). Therefore

$$\frac{\sigma^2 s_j^2 \Delta t}{2(\Delta K)^2} = \frac{(\Delta t/\Delta K)\mathrm{Cal}_0}{(\Delta K)\mathrm{Fly}_0}$$

Conclusion: at node (s_j, t_n), the local volatility calibrated at time 0 is

$$\sigma(s_j, t_n) = \sqrt{\frac{2}{s_j^2} \times \frac{\mathrm{Cal}_0(s_j, t_n)}{\mathrm{Fly}_0(s_j, t_n)}}$$

Conceptually, the local volatility $\sigma(s, t)$ can be inferred from prices of options with strikes near s and expiries near t.

Calibration of local volatility in a trinomial tree

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Comments:

- We have derived a discrete version of the "Dupire equation." We'll return to this formula in the PDE context.
- If you try to implement this formula directly, the calibrated σ will be sensitive to noise in the option price observations. In practice some regularization/smoothing procedure is advisable.
- Calibration is much easier in the case where σ depends only on t instead of jointly on (S, t).

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Skip

Another path-dependent option: Lookback

A fixed-strike lookback with strike K, start date 0, and expiry T pays

$$\text{Call:} \qquad (\max_{t\in[0,T]\cap\mathcal{T}} S_t - K)^+$$

$$\text{Put:} \qquad (K - \min_{t\in[0,T]\cap\mathcal{T}} S_t)^+$$

where $\mathcal{T}$ is some set of monitoring times.

Pricing a fixed-strike lookback call

With risk-neutral dynamics

$$\mathrm{d}S_t = rS_t\mathrm{d}t + \sigma S_t \mathrm{d}W_t,$$

the level of S_t determines the time-t conditional distribution of

$$\max_{u\in[t,T]\cap\mathcal{T}} S_u.$$

But that's not enough information, if option is on $\max_{u\in[0,T]\cap\mathcal{T}} S_u$.

Need also to track the running maximum

$$M_t := \max_{u\in[0,t]} S_u.$$

So the time-t price of a fixed-strike lookback call is a function of *three* state variables. Due to path dependence, not just a function of (t, S_t).

$$C(t, S_t, M_t)$$

Pricing a fixed-strike lookback call in a tree

- Recall: the knockout option, also path-dependent, has price

$$C(t, S_t, \mathbf{1}_{\text{knockout prior to time } t})$$

  Here the path-dependence is simple. The path-dependent state variable has only two states: 0/1 (live/dead).

  And the "1" state does not need to be tracked in the tree.

- Lookback is more complicated than the barrier. At each node (t, S_t) in the tree, need to store a table of values associating each possible M_t with $C(t, S_t, M_t)$.

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Example

Assumes r=0	
nu	-0.042
sigma	0.29
deltaX	0.25
deltaT	0.25

running max	option price: fixed strike lookback call

pu	0.15
pm	0.66
pd	0.19

K 100

		n=1 time 0	n=2 time 0.25	n=3 time 0.5	n=4 time 0.75	n=5 time 1.00
j=4	271.83					271.83 171.83
j=3	211.70				211.70 120.60	211.70 111.70
j=2	164.87			164.87 77.71	164.87 71.81	211.70 111.70 164.87 64.87
j=1	128.40		128.40 42.47	128.40 38.40	164.87 64.87 128.40 33.80	164.87 64.87 128.40 28.40
j=0	100	100.00 13.84	100.00 10.96	128.40 29.20 100.00 7.79	128.40 28.40 100.00 4.21	164.87 64.87 128.40 28.40 100.00 0.00
j=-1	77.88		100.00 1.57	100.00 0.62	128.40 28.40 100.00 0.00	128.40 28.40 100.00 0.00
j=-2	60.65			100.00 0.00	100.00 0.00	128.40 28.40 100.00 0.00
j=-3	47.24				100.00 0.00	100.00 0.00
j=-4	36.79					100.00 0.00